

SCHOOL OF COMPUTER SCIENCES
UNIVERSITI SAINS MALAYSIA

CDS503: Machine Learning

Semester 1, 2024/2025

FINAL REPORT

Group Number: G03

Project Title: Patient Risk Assessment & Resources Allocation

Name	Matric No.	USM Email Addresses	Experiment Set
Lee Jun Xian	23202396	junxian010729@student.usm.my	Comparing ML model
Lee Zi Ting	23202363	ziting.lee@student.usm.my	Features Selection
Liew Yu Xuan	23202456	yuxuan98@student.usm.my	Ensemble Learning

Task 2: Data Set Selection

2.1.	Data Source Diabetes 130-US hospitals for years 1999-2008 dataset is sourced from UCI Machine Learning Repository (UCI Machine Learning Repository: Diabetes 130-US hospitals for years 1999-2008 Data Set, 2023). URL Link: Diabetes 130-US Hospitals for Years 1999-2008
2.2.	Data Collection The dataset is collected from Diabetes 130 US hospitals and integrated delivery networks, collected over a period of 10 years from 1999 to 2008. Data were retrieved from the database for experiences accomplishing the specified criteria. (1) It is an inpatient encounter. (2) It is a diabetic encounter. (3) The length of stay was at least 1 day and at most 14 days. (4) Laboratory tests were performed during the encounter. (5) Medicines were administered during the encounter. The dataset comprises 101,766 instances and 50 attributes that represent patient and hospital outcomes.
2.3.	Data Attribute and Type Please refer to Appendix A for the data attributes, types, and descriptions within the dataset.

Task 3: Problem Framing

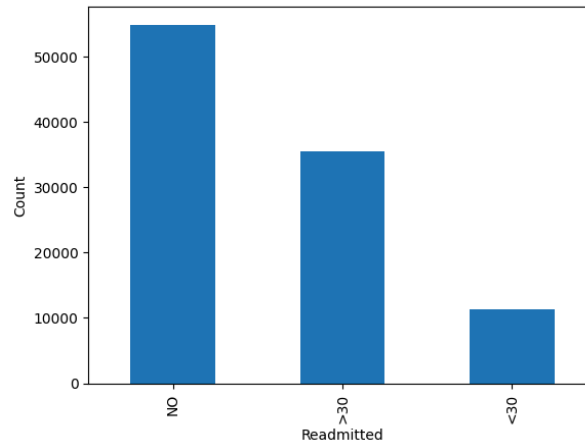
Hospital readmission poses significant challenges to the healthcare system. It is impacting the hospitals, patients and the health care systems. The high readmission rates can indicate the potential gaps in patient care and will lead to increased financial burdens or straining the hospital resources. Hospitals use readmission rates as a quality metric to assess the performance and improve patient outcomes via the comparative analysis with industry benchmarks.

To overcome these challenges, hospitals need to implement effective readmission reduction programs aimed at identifying and mitigating factors contributing to patient readmission. However, the success of such initiatives heavily depends on accurately identifying high risk patients and understanding the patterns that lead to readmissions. The failure of action can result in ineffective interventions and increased healthcare costs.

The primary goal is to develop a predictive model using supervised machine learning algorithms to classify patient readmission outcomes based on some personal health attributes and hospital related factors. The classification problem focuses on predicting these three outcomes.

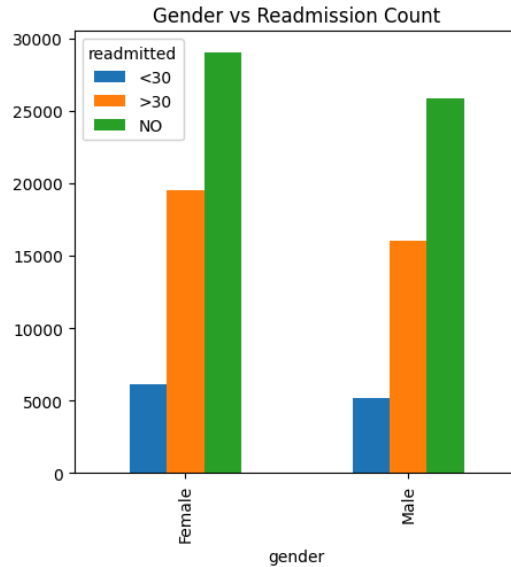
- Not Readmitted (NO): Patients who did not follow up visit hospital
- Readmitted in Less than 30 days (<30): Patients who readmitted within 30 days
- Readmitted in More than 30 days (>30): Patients who readmitted after 30 days.

Task 4: Data Preparation

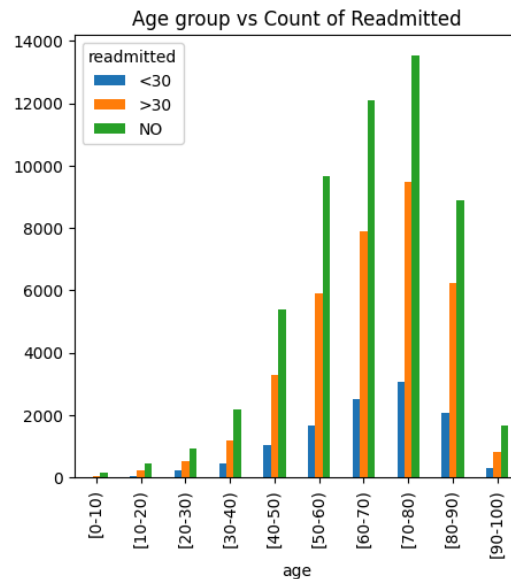
4.1.	Exploratory Data Analysis																			
4.1.1.	Basic Data Description and Characteristics The dataset comprises 101,766 instances, featuring 49 attributes and 1 labelled data. There are two data types which contain 41 categorical and 8 numerical among the features. The labelled data in column 'readmitted' exhibits imbalanced multinomial classes where NO contains 54,864 instances, > 30 contains 35,545 instances, and < 30 contains 11,357 instances. The readmission count is illustrated in Figure 1. Readmission Counts <table><thead><tr><th>Readmitted</th><th>Count</th></tr></thead><tbody><tr><td>NO</td><td>54864</td></tr><tr><td>>30</td><td>35545</td></tr><tr><td><30</td><td>11357</td></tr></tbody></table> Figure 1: Labelled Data Distribution	Readmitted	Count	NO	54864	>30	35545	<30	11357											
Readmitted	Count																			
NO	54864																			
>30	35545																			
<30	11357																			
4.1.2.	Detect Aberrant & Missing Values Three attributes are illustrated in Figure 2, specifically 'weight', 'medical_specialty' and 'payer_code' which exhibit significant missing values with rates of 96%, 49%, and 39.5%, respectively. Consequently, these were dropped from further investigation. <table><thead><tr><th></th><th>Total</th><th>Percent</th></tr></thead><tbody><tr><td>weight</td><td>98569</td><td>0.968585</td></tr><tr><td>max_glu_serum</td><td>96420</td><td>0.947468</td></tr><tr><td>A1Cresult</td><td>84748</td><td>0.832773</td></tr><tr><td>medical_specialty</td><td>49949</td><td>0.490822</td></tr><tr><td>payer_code</td><td>40256</td><td>0.395574</td></tr></tbody></table> Figure 2: Missing Value Percentage		Total	Percent	weight	98569	0.968585	max_glu_serum	96420	0.947468	A1Cresult	84748	0.832773	medical_specialty	49949	0.490822	payer_code	40256	0.395574	
	Total	Percent																		
weight	98569	0.968585																		
max_glu_serum	96420	0.947468																		
A1Cresult	84748	0.832773																		
medical_specialty	49949	0.490822																		
payer_code	40256	0.395574																		

4.1.3. Demographic Analysis

The demographic distribution indicates that 53.76% of the population is female, while 46.23% is male with only three instances of unknown gender. The distribution pattern of each label class exhibits similarity across genders, as illustrated in Figure 3. The dataset is primarily composed of individuals who are aged between 60 and 90 years, representing the elderly population.



readmitted	<30	>30	NO
gender			
Female	6152	19518	29038
Male	5205	16027	25823



readmitted	<30	>30	NO
age			
[0-10)	3	26	132
[10-20)	40	224	427
[20-30)	236	510	911
[30-40)	424	1187	2164
[40-50)	1027	3278	5380
[50-60)	1668	5917	9671
[60-70)	2502	7897	12083
[70-80)	3069	9475	13522
[80-90)	2078	6223	8896
[90-100)	310	808	1675

Figure 3: Demographic Analysis

4.1.4. Categorical Features Exploration

Asymmetric distribution is observed for various features including Admission Type, Discharge Disposition Type and Admission Sources Type. The respective predominants contributed over 50%, as illustrated in Figure 4.

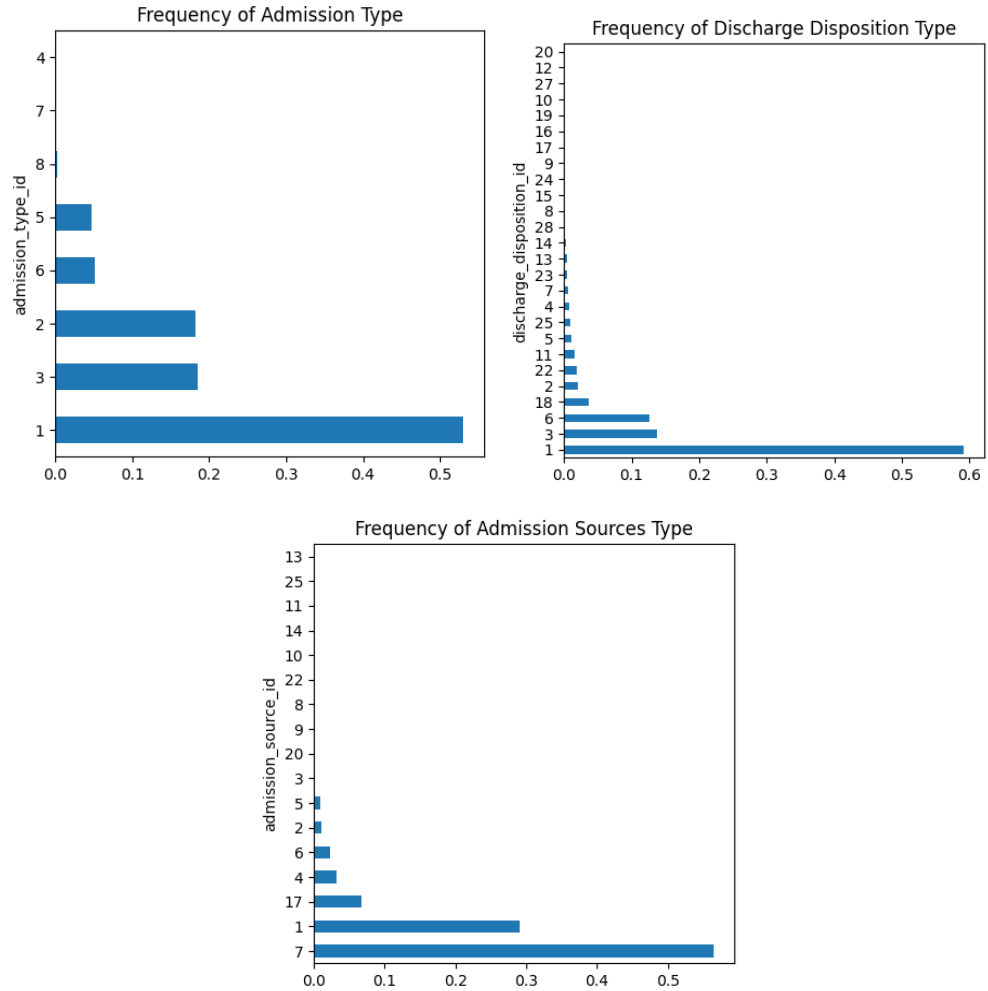


Figure 4: Admission and Discharge's Profile

The number of diagnoses per patient is included among the features, with a significant proportion diagnosed with multiple medical conditions, reaching up to 16. Figure 5 illustrates that the readmission rate was significantly concentrated among patients with 5 to 9 medical conditions.

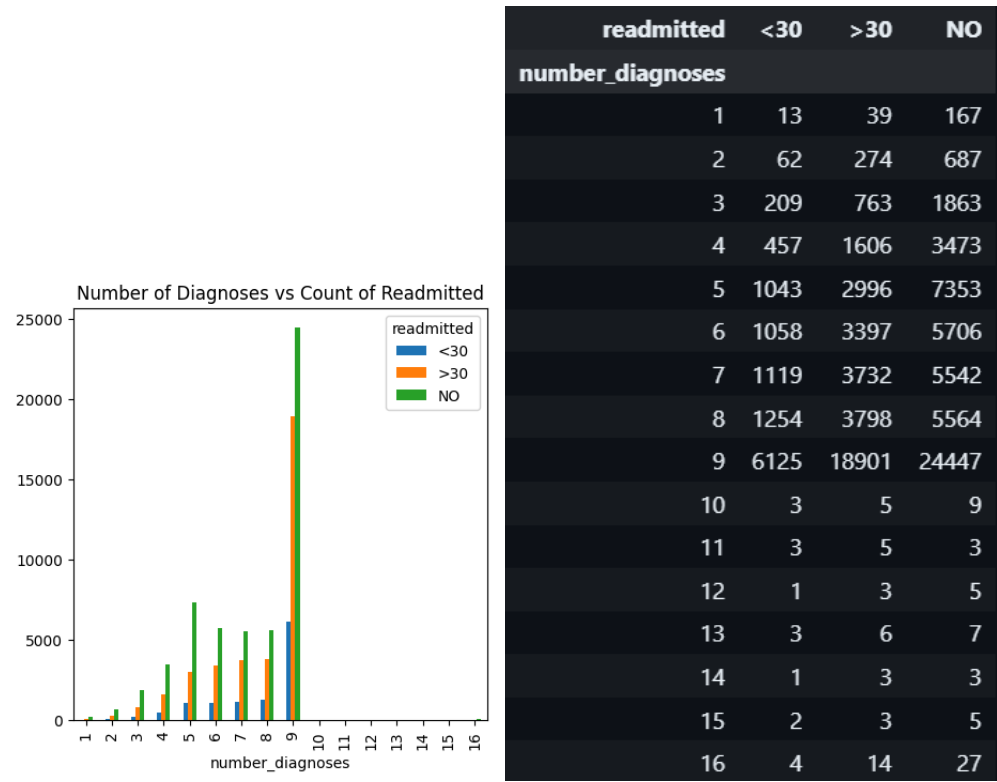


Figure 5: Number of Diagnoses

Over 50% of the samples were prescribed diabetes medication. The dataset contains 22 types of medicine dosage variations that can be categorized and illustrated through adjusted features. The designation “No” is applied when dosages are recorded as “No” and/or “Steady.” Conversely, “Ch” is indicated for “Up” and “Down” if applicable. Figure 6 shows an equal distribution between change and no change.



Figure 6: Medicine Prescription and Changes

4.1.5. Numerical Features Exploration

We analyzed the multicollinearity for numerical data types. The heatmap coefficient, all below 0.5, indicates that these features are weakly correlated, indicating a weak tendency for them to balance each other out. They can provide unique information.

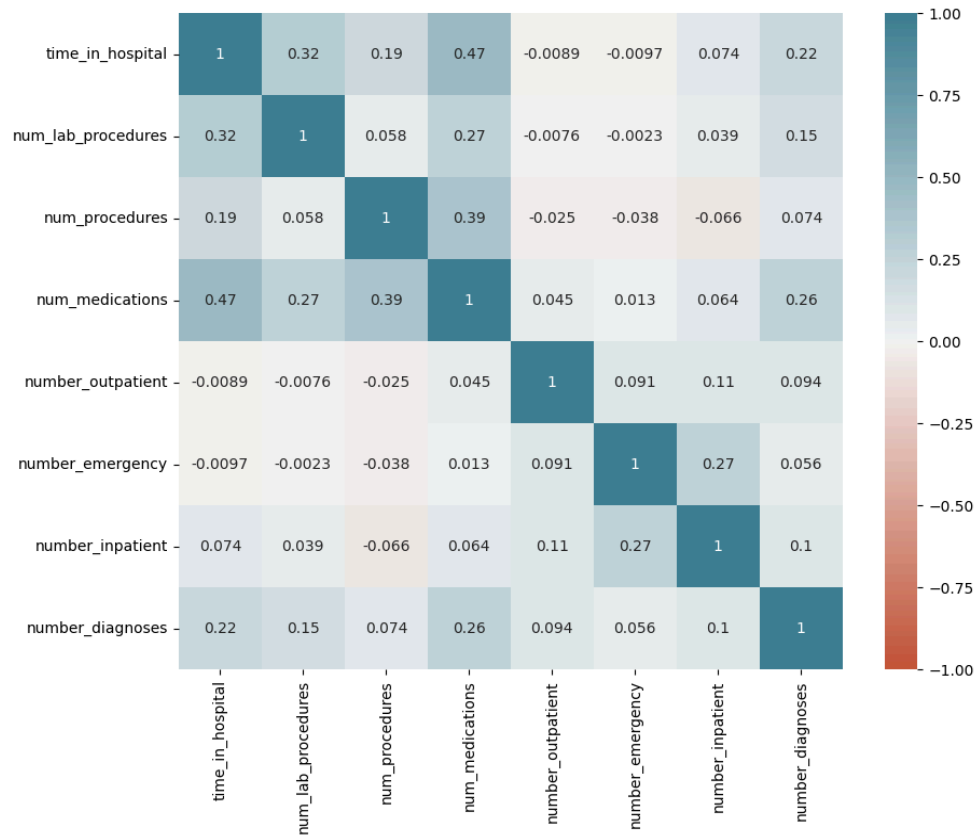


Figure 7: Heatmap Coefficient for Numerical Features

4.2.	Data Preprocessing It is important to remove irrelevant data and preprocess the dataset into an appropriate format prior to fitting it into the machine learning model. The analysis of missing values (4.1.2) indicates that "weight," "payer_code," and "medical_specialty" exhibit the greatest number of missing data. Consequently, these variables which are along with unique id columns such as "encounter_id" and "patient_nbr," are excluded from the dataset. Additionally, 22 types of medicinal information were excluded due to the majority of a single value distribution. Missing categorical values for minority categories are computed using the mode. Given that "discharge_disposition_id" and "admission_source_id" indicate significant binning, the inaccuracy of the indexing demands index replacement. "diag_1," "diag_2," and "diag_3" are represented using ICD-9 codes, which combine numerical and textual formats. It provides the grouping and mapping of codes according to disease categories. Winsorization is applied on numerical features, specifically "num_medications," "num_lab_procedures," "num_procedures," and "number_diagnoses," to substitute extreme values with the upper and lower boundaries established by the interquartile range, thereby reducing the influence of outliers. Binning is applied to "number_inpatient" and "number_outpatient" to discretize the values. Overall, all categorical data with a greater number of labels is mapped and subjected to one-hot encoding, except in Experiment 2, to address the ranking issue associated with numerical data when applying label encoding. Label encoding is applied to categorical data with a limited number of labels. The initial 50 feature columns were reduced to 23, resulting in an adjustment to 94 feature columns, which includes one-hot encoded columns, as part of the cleaned data.
------	---

Task 5: Experiment Setup

Experiment 1: Comparing Machine Learning Algorithms

Classification is completed on 4 different algorithms to identify the best machine learning classification model for predicting the need of readmission based on patient personal health information within the supervised machine learning in experiment 1. To define the best model, F1-score will be used as the metric due to the dataset's imbalance. These models will serve as a baseline or reference for experiment 2 and experiment 3. It is to determine any improvements since experiment 1 used the full preprocessed dataset for the training and testing purposes.

The classification algorithms used for Experiment 1 are listed below:

- Dummy Classifier
- Decision Tree
- K-Nearest Neighbour (KNN)
- Naive Bayes (Multinomial, Bernoulli, Gaussian)

The dataset is splitted into a training validation set and a test set. Within the dataset, 80% will be used for training validation and 30% will be used as the test dataset. The best model is defined based on validation scoring during the experiment training. After evaluating the baseline performance of various machine learning models, the model that achieved the highest F1-Score is Multinomial Naive Bayes. It is selected for further optimization. Hyperparameter tuning will then be conducted to enhance its accuracy and overall performance. For validation purposes, 10-fold cross-validation is applied, partitioning the dataset into 10 subsets, with each iteration utilizing 9 subsets for training and the remaining 1 subset for validation. This process is repeated across all subsets to ensure a robust evaluation of the model's performance. The train-validation result is shown below.

Classifier	Accuracy	Weighted Precision	Weighted Recall	Weighted F1
Dummy	0.5403	0.2919	0.5403	0.3790
Decision Tree	0.4723	0.4771	0.4723	0.4746
K Nearest Neighbor	0.4927	0.4709	0.4927	0.4802
Multinomial Naive Bayes	0.5759	0.5344	0.5759	0.5352
Bernoulli Naive Bayes	0.5722	0.5318	0.5722	0.5332
Gaussian Naive Bayes	0.1976	0.6599	0.1976	0.1703

Table 1: Validation Performance in Experiment 1

According to Table 1, the baseline model evaluation was conducted using various machine learning algorithms, including dummy classifiers, decision trees, k nearest neighbor and naive bayes. The performance of each model was measured based on key metrics such as precision, recall, F1-score and accuracy. Among the evaluated models, the Multinomial Naive Bayes achieved the highest F1-Score of 0.5352, and it is the best performing model in terms of overall predictive accuracy as Multinomial NB is specifically designed for categorical and count-based data. While decision trees and KNN might struggle with high dimensional categorical data leading to lower overall performance. For Gaussian NB, it assumes that the features follow normal distribution and it might not be well-suited for the dataset. The dataset consists of categorical features which do not exhibit a Gaussian distribution. This mismatch leads to poor probability estimations and incorrect classification.

Best Classifier	Parameters	F1-Score
Multinomial Naive Bayes	{'alpha': 1.0, 'fit_prior': True}	0.5321

Table 2: Hyperparameter Tuning for Best Validation Performance in Experiment 1

As shown in Table 2, hyperparameter tuning is performed on the Multinomial Naive Bayes model to further optimize the performance. The tuning process involved adjusting the key parameters like alpha and fit prior to identify the optimal combination for improving predictive accuracy. However, the best F1-Score on validation data achieved after tuning is 0.5321 which is slightly lower than the baseline F1-Score of 0.5352 from Table 1. The best performing parameters identified through tuning were ‘alpha:1.0’ and ‘fit_prior:True’.

Class	Precision	Recall	F1-Score	Support
0 (<30)	0.2735	0.0611	0.0999	2241
1 (>30)	0.4921	0.3512	0.4099	7116
2 (NO)	0.6149	0.8261	0.7050	10997

Table 3: Classification Report of Test Data

Metric	Score
Test Accuracy	0.5759
Macro Average F1-Score	0.4049
Weighted Average F1-Score	0.5352
Test F1-Score	0.5759

Table 4: Overall Test Performance in Experiment 1

Table 3 shows the classification report of test data and Table 4 shows the overall test performance after hyperparameter tuning in Experiment 1. The dataset shows a significant class imbalance with class 2 (NO) having the highest number of samples of 10997. It has led to the highest recall of 0.8261 and F1-score of 0.7050. While for the class 0 (<30) and 1 (>30) which have fewer samples, they have lower recall values of 0.0611 and 0.3512 respectively. The imbalanced data allows the model to perform better for class 2 due to being more sensitive to data as it is the majority class in the dataset where the model to learn from, while for the minority class 0 & 1 struggling in classification due to the limited training samples.

In an imbalance dataset, accuracy is not reliable as it only counts the correct predictions from all the classes equally. Instead of accuracy, F1-score is a better way to evaluate the model performance as it balances the precision and recall which is giving a better evaluation across all classes. In binary classification, the accuracy is calculated or divided based on 2 classes. However, the performance will be averaged across three or more classes in multiclass classification. This will result in naturally low accuracy as it considers more classes and leads to difficulty to achieve a high score. Hence, the result in Multinomial Naive Bayes's performance is better in the majority class compared to the minority classes and leads to lower F1-score.

Experiment 2: Feature Selecting

By implementing feature selection, it is to find out the way to reduce the dimensionality to mitigate the overfitting issues and improve the performance by removing the mislead features as well as enhancing the computation time on the training process. In this experiment, there are few features selection methods such as Chi-Squared, ANOVA, Tree-Based Feature Importance, and Correlation Analysis that have been selected to assist the selection of the features.

Chi-Squared

After data preprocessing, the new clean dataset is stored in a csv file named “diabetic_data_without_outliers.csv”. The new dataset variables contain boolean values (true and false) and one-hot encoder classification labels (0, ...) . The process begins by identifying and extracting classification features (including objects, categories, and Boolean data types) from the training dataset. These features are then converted to a numeric format so that new binary columns are created for each category value for each feature. Next, the top 10 features are selected for training with different classifiers (including Decision Tree, KNN, Gaussian NB, Bernoulli NB, and Multinomial NB) by setting “SelectKBest” using chi2 and a score function with $k = 10$ in order to evaluate the classifiers with different selected features. Below are the top 10 features that were selected with the chi2 score.

	Features	Chi2 Score
1	DD-Expired	1110.079673
2	DD-Another_RehadFac	406.458495
3	DD-Home_HealthService	289.173942
4	AS-Trans_Another_HealthCareFaculty	289.117256
5	AS-Trans_Hosp	247.471593
6	diag1_Neoplasms	201.577666
7	AS-EmergencyRoom	188.760252
8	AT-Elective	173.614048
9	DD-Hospice_MedicalFacility	171.093423
10	DD-Home	165.024400

Table 5: Top 10 Features Selected with Chi2 Score

From Table 6, The performance of the classifiers varied widely based on the results of the chi-square feature selection. The Decision Tree, Bernoulli NB, and Multinomial NB classifiers perform relatively similarly in terms of accuracy, recall, and F1-score metrics, with accuracies ranging from about 0.541 to 0.542, weighted recalls in the same range, and F1-scores from about 0.42 to 0.44. KNN performed slightly worse with an accuracy of 0.4582, a weighted recall of 0.4582 and an F1-score of 0.4480. Gaussian Plain Bayes performed the worst, with an accuracy and weighted recall of 0.3741, along with a minimum F1-score of 0.2668 and a slightly higher weighted precision of 0.6137. Overall, these results highlight that using Chi-Squared feature selection can impact different classification models differently, with Naive Bayes not performing very well with the selected categorical features in comparison with other models like Decision Tree.

Classifiers	Accuracy	Weighted Precision	Weighted Recall	Weighted F1
Decision Tree	0.5411	0.4662	0.5411	0.4388
KNN	0.4582	0.4594	0.4582	0.4480
Gaussian NB	0.3741	0.6137	0.3741	0.2668
Bernoulli NB	0.5422	0.4509	0.5422	0.4250
Multinomial NB	0.5422	0.4509	0.5422	0.4250

Table 6: Validate Result of Selected Features from Chi-Squared

ANOVA

The ANOVA feature selection is used to evaluate the relationship between numerical features and the target variable in a classification problem. First, the code identifies all numerical features within the training data. These features are then subjected to the ANOVA test using SelectKBest with `f_classif` as the scoring function, selecting the top 10 features based on the computed F-statistics, which is used to quantify the variance between the different category means of each feature. Below are the top 10 features that were selected with the ANOVA score.

	Features	ANOVA Score
1	number_inpatient	2516.113679
2	number_diagnoses	535.631004
3	number_outpatient	465.873155
4	number_emergency	449.762693
5	diabetesMed	163.537837
6	num_medications	158.891203
7	time_in_hospital	147.530859
8	change	94.894415
9	num_procedures	92.800565
10	num_lab_procedures	70.526934

Table 7: Top 10 Features Selected with ANOVA Score

From Table 8, the result shows that of all classifiers, the Decision Tree performs the worst with accuracy, weighted recall and a weighted f1 score of about 0.44. The KNN performs better with a metric of about 0.48, while the Gaussian NB, Bernoulli NB, and Multinomial NB all achieve very similar results, with accuracy and weighted recall of about 0.56 and f1 score of about 0.51. The ANOVA approach resulted in a much better performance of the Gaussian Naive Bayes algorithm, and the selected numerical features may be more useful for this algorithm. In addition, the performance of the Bernoulli and Multinomial Naive Bayes algorithms remained very similar with this different feature selection method.

Classifiers	Accuracy	Weighted Precision	Weighted Recall	Weighted F1
Decision Tree	0.4440	0.4544	0.4440	0.4488
KNN	0.4855	0.4655	0.4855	0.4739
Gaussian NB	0.5563	0.5111	0.5563	0.4967
Bernoulli NB	0.5636	0.4870	0.5636	0.5155
Multinomial NB	0.5646	0.5178	0.5646	0.5143

Table 8: Validate Result of Selected Features from ANOVA

Tree-Based Feature Importance

The Tree-Based Feature Importance method was utilized to identify the most informative numerical features by leveraging the intrinsic feature selection capability of decision trees. The process begins by fitting a decision tree classifier to the digital features in the training data. The decision tree model learns to make predictions by splitting the data based on the most relevant features. Feature importance is extracted from the fitted model and sorted in descending order to reflect the relative importance of each feature. The top 10 features are selected based on the importance score.

	Features	Tree-Based Importance
1	num_lab_procedures	0.247285
2	num_medications	0.182769
3	time_in_hospital	0.120199
4	age	0.081203
5	num_procedures	0.066315
6	number_diagnoses	0.063986
7	insulin	0.054547
8	number_inpatient	0.038637
9	A1Cresult	0.033306
10	gender	0.032766

Table 9: Top 10 Features Selected with Tree-Based Importance

From Table 10, the performance results show that the Decision Tree and KNN models perform the worst, with accuracy and weighted recall ranging from 0.45 to 0.48. The Bernoulli NB and Multinomial NB perform similarly, with accuracy and weighted recall around 0.56. The Gaussian NB outperforms the first two models with an accuracy and weighted recall of about 0.56, but still has the worst F1 score of about 0.49. These results emphasize the difference between how the models work with the data selected by the Tree-Based method, which seems to favor Naive Bayes algorithms.

Classifiers	Accuracy	Weighted Precision	Weighted Recall	Weighted F1
Decision Tree	0.4509	0.4595	0.4509	0.4549
KNN	0.4866	0.4669	0.4866	0.4754
Gaussian NB	0.5609	0.5164	0.5609	0.4926
Bernoulli NB	0.5602	0.4828	0.5602	0.5102
Multinomial NB	0.5683	0.5260	0.5683	0.5100

Table 8: Validate Result of Selected Features from Tree-Based Feature Importance

Correlation Analysis

The correlation-based feature selection method aims to identify the features that are most linearly correlated with the target variable. The process begins by coding the target variable as a numeric value using category codes. Then, a correlation matrix is computed to determine the correlation between all pairs of features, including the target variable. The higher the absolute value of the correlation coefficient, the stronger the linear relationship. The correlation between each feature and the target variable is sorted in descending order, and then the top 10 features with the highest correlation are selected.

	Features	Correlated Features
1	DD-Expired	0.107172
2	DD-Home	0.051492
3	AS-Trans_Another_HealthCareFaculty	0.046176
4	AS-Trans_Hosp	0.044082
5	change	0.042504
6	AT-Elective	0.041920
7	num_procedures	0.038360
8	diag1_Neoplasms	0.037659
9	DD-Hospice_MedicalFacility	0.037599
10	DD-Hospice_Home	0.035495

Table 11: Top 10 Features Selected with Correlated Features

From Table 12, the result shows that the decision tree achieved an accuracy and weighted recall of approximately 0.54, with an F1-score of 0.379. The KNN model performance was much worse, with an accuracy and weighted recall of about 0.44 and an F1-score of 0.429. The Gaussian NB is the worst performing of all, with an accuracy and recall of approximately 0.37 and a very low F1-score of about 0.26. Bernoulli and Multinomial Naive Bayes algorithms achieve the exact same results, with an accuracy and recall of about 0.54, and an F1-score of 0.379. This method also has a tendency to favor Bernoulli and Multinomial, which could indicate similarities between these two models and also with the feature set selected by this method. The results demonstrate that the selected features by the correlation method favor the decision tree in comparison with KNN and that Gaussian Naive Bayes is not suited to features selected by this method.

Classifiers	Accuracy	Weighted Precision	Weighted Recall	Weighted F1
Decision Tree	0.5400	0.4360	0.5400	0.3798
KNN	0.4459	0.4641	0.4459	0.4293
Gaussian NB	0.3773	0.6039	0.3773	0.2683
Bernoulli NB	0.5403	0.2919	0.5403	0.3790
Multinomial NB	0.5403	0.2919	0.5403	0.3790

Table 12: Validate Result of Selected Features from Correlation Analysis

After evaluating the performance of the classifiers with different feature selections, ANOVA was found to be the most effective approach for this dataset and classification task. While Chi-Squared proved useful for Decision Trees, and Tree-Based Importance and Correlation Analyses highlighted specific models, these methods did not achieve the consistent and comprehensive performance observed in ANOVA. The Naive Bayes classifiers also had good performance, consistently achieving the highest accuracy and recall scores with the selected features from the ANOVA method. While other methods had some classifiers that performed well, the consistent performance of the Naive Bayes classifiers using the ANOVA feature sets indicates that these features provide the most suitable input for overall model performance.

Experiment 3: Ensemble Learning

In this experiment, ensemble learning is implemented to improve the model performance by combining multiple models to form a robust ensemble model. There are 3 ensemble learnings involved in the experiment, which are Bagging, Boosting and Voting.

Bagging

The code provided explores the impact of the ensemble learning technique ‘bagging’ on the performance of five different base classifiers: decision trees, KNN, Gaussian Naïve Bayes, Bernoulli Naïve Bayes and Multinomial Naïve Bayes. In each case, the BaggingClassifier was instantiated using 50 base estimators (`n_estimators=50`) and a random set of states for reproducibility. This process involves creating multiple subsets of training data using random sampling with replacement (bootstrapping) and training a base classifier on each subset. When making predictions, each base classifier contributes one vote, and the final prediction is based on the majority vote.

Classifiers	Accuracy	Weighted Precision	Weighted Recall	Weighted F1
Decision Tree	0.53	0.50	0.53	0.51
KNN	0.52	0.47	0.52	0.49
Gaussian NB	0.14	0.64	0.14	0.07
Bernoulli NB	0.54	0.52	0.54	0.43
Multinomial NB	0.57	0.53	0.57	0.52

Table 13: Bagging Results for each classifiers

The results demonstrate varying effectiveness of bagging across the different base classifiers. The Bagging Classifier using a Decision Tree base classifier had an accuracy of 0.53 and an F1-score of 0.51. The Bagging Classifier using a KNN base classifier also performed worse, with an accuracy of 0.52 and an F1-score of 0.49. Bagging with Gaussian Naive Bayes had by far the worst performance, with an accuracy of 0.14 and an F1-score of 0.07, with a very low weighted recall of 0.14. However, when GaussianNB was used in this way, the results were very imbalanced, with high precision for the prediction of class 2. Bagging with BernoulliNB had a slightly higher accuracy than the previous two classifiers at 0.54, and an F1-score of 0.43. The best model was Bagging with MultinomialNB, which achieved an accuracy of 0.57 and the highest F1-score at 0.52, which is a great improvement over all of the other bagging models, and especially the GaussianNB model. These results highlight that the effectiveness of bagging

highly dependent on the choice of base estimator and that it is not guaranteed to improve performance. While bagging improved the performance of the MultinomialNB model, the results with the GaussianNB model were terrible. Furthermore, the results show a significant performance boost with using the BernoulliNB and MultinomialNB models with this method.

Boosting

The provided code investigates the effectiveness of AdaBoost, a boosting ensemble learning method, on three base classifiers: Decision Tree, Bernoulli Naive Bayes, and Multinomial Naive Bayes. In each experiment, an AdaBoostClassifier is constructed with 50 base estimators and a fixed random state for reproducibility. AdaBoost operates by sequentially training these base estimators, with each subsequent estimator focusing on the instances that were misclassified by the previous ones. This iterative process gives more weight to difficult samples, effectively making the model better at classifying them. The final prediction is a weighted vote, giving higher importance to estimators that performed better during the training process.

Classifiers	Accuracy	Weighted Precision	Weighted Recall	Weighted F1
Decision Tree	0.46	0.46	0.46	0.46
Bernoulli NB	0.57	0.49	0.57	0.48
Multinomial NB	0.57	0.48	0.57	0.49

Table 14: Boosting Results for each classifiers

The results show a variation in how AdaBoost affects these classifiers. AdaBoost, when used with a Decision Tree as the base classifier, performed the worst, achieving an accuracy and F1-score of 0.46. When AdaBoost is combined with BernoulliNB, it gives a better accuracy of 0.57, but the model strongly favors the final class and achieves a very low recall for the other classes, with an F1-score of 0.48. AdaBoost combined with MultinomialNB also achieved an accuracy of 0.57, with an F1-score of 0.49. However, the same phenomenon as in the BernoulliNB classifier is observed, where the model has a very high recall for the final class but very low recall for the other classes. This implies that both the Bernoulli and Multinomial Naive Bayes classifiers tend to overfit on the final class when combined with boosting.

The code intentionally excludes KNN and Gaussian Naive Bayes from the boosting experiments because these algorithms are often incompatible with boosting. KNN, being a distance-based algorithm, does not have a natural weighting mechanism that boosting algorithms use to emphasize more informative models. In addition, because KNN is computationally expensive, boosting would make the algorithm run incredibly slowly. Gaussian Naive Bayes, which assumes

feature independence, also does not work well with boosting. Boosting emphasizes correcting past mistakes, which creates dependencies between subsequent models. This dependence is contrary to the independence assumption of Gaussian Naive Bayes, which is why they do not perform well when boosted. Ultimately, the AdaBoost algorithm is best suited for models that can be easily adjusted and weighted during the boosting process and that have room for improvement, which is why tree-based algorithms and discrete Naive Bayes algorithms are more suitable for boosting than other methods.

Voting

The provided code explores the impact of the Voting Classifier, an ensemble learning technique that combines the predictions of multiple different models. In this experiment, five diverse classifiers are used: Decision Tree, KNN, Gaussian Naive Bayes, Bernoulli Naive Bayes, and Multinomial Naive Bayes. Two types of voting are performed: hard voting and soft voting. Hard voting, specified by `voting='hard'`, assigns the final class label based on the majority vote among the different base classifiers, i.e., whichever class is predicted most often by the classifiers. On the other hand, soft voting, specified by `voting='soft'`, calculates a weighted average of the predicted class probabilities provided by each classifier. These weights are typically determined by the training performance of the classifiers. The class with the highest weighted average probability is chosen as the prediction. Both types of voting are evaluated on the testing set using accuracy and classification reports.

Classifiers	Accuracy	Weighted Precision	Weighted Recall	Weighted F1
Hard	0.51	0.5	0.51	0.49
Soft	0.5	0.5	0.5	0.48

Table 15: Voting Results

The results for hard voting reveal a model that achieves an accuracy of 0.51 and an F1-score of 0.49. For soft voting, the results reveal an accuracy of 0.50 and an F1-score of 0.48. These results show that the two voting methods performed very similarly for this task, with very small variations between their performance metrics. Both also achieved similar results with the weighted averages of precision, recall, and F1-score, suggesting that the two methods are performing similarly at the class level and not just at the overall accuracy level. Given how similar the results are with both soft and hard voting, this may indicate that the models are likely very similar to each other and are not very diverse. It is also possible that the weights that soft voting used were very similar for all of the models, which would cause the soft voting model to become very similar to the hard voting model. Overall, these results show that the voting

approach does not lead to better results in comparison with the Bagging and Boosting approaches discussed earlier, which suggests that perhaps these approaches are better suited to this dataset. The results also indicate that the two types of voting used in this approach give very similar results.

After examining the results of bagging, boosting, and voting ensemble methods, the boosting method, specifically with either Bernoulli Naive Bayes or Multinomial Naive Bayes as the base classifier, appears to be the most efficient for this specific dataset. Although both bagging and voting showed some degree of improvement over individual classifiers, the accuracy and F1-scores achieved with those methods were generally lower than the best models obtained with boosting. While bagging with MultinomialNB achieved a reasonable accuracy and F1-score, boosting with BernoulliNB and MultinomialNB provided similar results and tended to have higher recalls and F1-scores in comparison with the other models when averaged. While the individual accuracy scores were often in the 0.5 to 0.57 range for the voting and bagging methods, the F1-scores were relatively lower and more inconsistent. The boosting approach using either BernoulliNB or MultinomialNB classifiers yielded the best performance overall and the highest weighted recall scores. In contrast, the decision tree model using boosting produced relatively bad results, and KNN and GaussianNB were not compatible with boosting. Additionally, although Gaussian Naive Bayes performed very poorly with bagging, the remaining models also had inferior performance in comparison with boosting with Bernoulli and MultinomialNB. Therefore, while the bagging and voting methods did provide some performance gain in comparison with base models, the boosting approach is the most efficient for this dataset due to its ability to focus learning on misclassified instances, which allows it to achieve the best performance.

Conclusion

In conclusion, there are various types of supervised machine learning algorithms to predict patient readmission outcomes based on personal health attributes and hospital-related factors. The classification task aimed to categorize patients into 3 groups: not readmitted, readmitted within 30 days, and readmitted after 30 days. The overall goal is to identify the best-performing model to support hospitals in implementing effective readmission reduction programs and optimizing resource allocation.

In experiment 1, the comparison of multiple machine learning algorithms, including the Dummy Classifier, Decision Tree, KNN and Naive Bayes. The results showed that the Multinomial Naive Bayes model achieved the highest performance with a weighted F1-score of 0.5352. Hyperparameter tuning was conducted to further optimize the overall performance. However, the result showed a slight decrease in validation F1-score to 0.5321. The evaluation highlighted the impact of data imbalance, with class 2 (NO) achieving the highest recall and F1-score due to the larger representation of the majority class, while minority classes 0 and 1 were lower in recall as the limited training samples.

In experiment 2, the feature selection techniques are explored to enhance the model performance by reducing dimensionality and mitigating the overfitting issues. Several feature selection methods, including Chi-Squared, ANOVA, Tree-Bases Feature Importance and Correlation Analysis are applied. The results showed the ANOVA-based feature selection has the best performance across different models, and this has benefited the Naive Bayes classifiers. This experiment confirmed that selecting the right features is crucial for improving the model performance and reducing the computational complexity.

In experiment 3, the ensemble learning techniques such as Bagging, Boosting and Voting are implemented to further improve the model performance. The result showed the Boosting technique, especially when applied to Bernoulli and Multinomial Naive Bayes classifiers. It achieved the highest performance in terms of accuracy and F1-score. The ensemble methods showed the potential of combining multiple models to achieve better predictive performance in handling imbalance data.

Overall, the selection of the appropriate machine learning model, feature selection methods and ensemble techniques is crucial for improving the predictive accuracy in an imbalanced dataset. The Multinomial Naive Bayes model with ANOVA-selected features and the Boosting ensemble technique showed the best results but is still not as promising as the imbalanced class distribution in the dataset. For future work, implement class weighting and data resampling methods to address the imbalanced class distribution issues to further enhance the model performance.

Appendix

Appendix A: Dataset Description

	Columns	Description	Target / Features	Attribute Type
1	encounter_id	Unique identifier of encounter	Features	Categorical
2	patient_nbr	Unique identifier for patient	Features	Categorical
3	race	Categorization of patient	Features	Categorical
4	gender	Categorization of female, male, unknown/invalid	Features	Categorical
5	age	Grouped in 10-years intervals	Features	Categorical
6	weight	Physical weight of patient	Features	Categorical
7	admission_type_id	Label for admission type	Features	Categorical
8	discharge_disposition_id	Label for discharge type	Features	Categorical
9	admission_source_id	Label for admission status type	Features	Categorical
10	time_in_hospital	Number of days between admission until discharge	Features	Numerical
11	payer_code	Label for payment methods	Features	Categorical
12	medical_specialty	Type of medical specialty	Features	Categorical
13	num_lab_procedures	Number of lab test during encounter	Features	Numerical
14	num_procedure	Number of procedures during encounter	Features	Numerical
15	num_medications	Number of distinct generic name administered during encounter	Features	Numerical
16	number_outpatient	Number of outpatients visit patient	Features	Numerical
17	number_emergency	Number of emergency visit patient	Features	Numerical
18	number_inpatient	Number of inpatient visit patient in year preceding encounter	Features	Numerical

19	diag_1	Primary diagnosis	Features	Categorical
20	diag_2	Secondary diagnosis	Features	Categorical
21	diag_3	Additional secondary diagnoses	Features	Categorical
22	number_diagnosis	Number of diagnoses entered in system	Features	Numerical
23	max_glu_serum	Indicate range of result for glucose serum test	Features	Categorical
24	A1Cresult	Indicate range of result for A1 c test	Features	Categorical
25	Medicines [From columns 25- 47 (Column Y-AU) in dataset]	Indicates if the dosage was prescribed or any dosage change (“No”, “Up”, “Down”, “Steady”)	Features	Categorical
26	change	Indicates whether any change in diabetes medication	Features	Categorical
27	diabetesMed	Indicates whether any diabetes medication prescribed	Features	Categorical
28	readmitted	Indicates whether patient readmitted for <30, >30 days or no readmission	Target	Categorical